Noisy Future Knowledge

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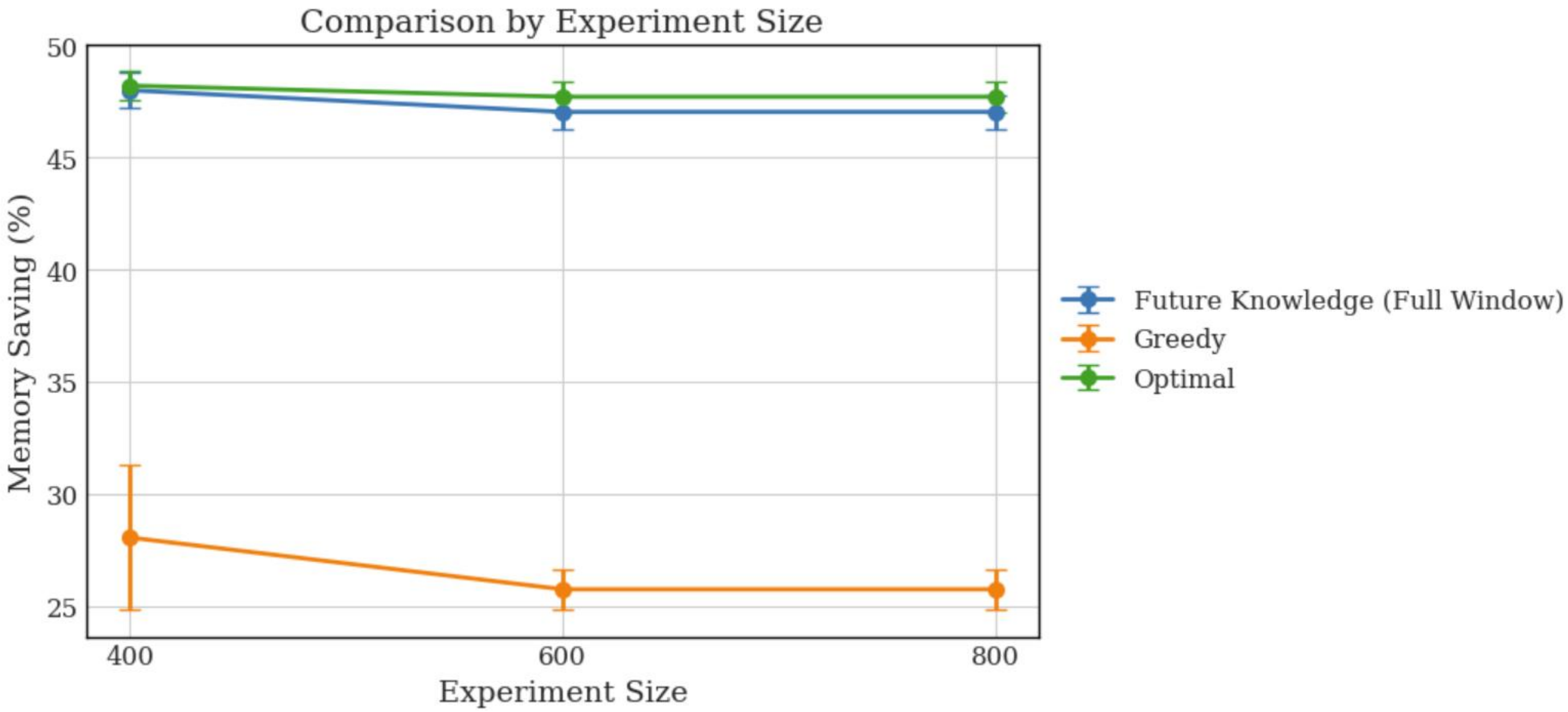
Motivation

Last time we saw that we can do well with perfect knowledge of the future and no migration (each experiment size value of 1 is equivalent to 5 minutes):



But what if the prediction is noisy?

Improved Sliding Window (Future Knowledge, No Migration)

This does not change the results when the sliding window size is the whole trace duration (no effective sliding). I also added multi-threading to make these faster.

1. Use the first *m* timesteps of the VM trace
2. At time t, have perfect knowledge *k* steps ahead
   - Fully covers the interval [t, t+k-1]
   - Know arrivals, departures, demands
3. At time *t*, solve linear program for allocations within the window
   - Lock in the placements ***only*** for the earliest VM arrival(s) within the window (*cannot* be changed later)
4. Advance time to *t+1* and repeat
5. In the end, measure the memory savings (based on the worst device peak) relative to the scenario without CXL pooling

Noisy Future Knowledge

1. Given the non-noisy future knowledge allocation for splitting memory across devices, we perturb it
2. Let the noise level be 10% (maximum)
3. Multiply each component in the allocation by a random value in [0.9, 1.1]
4. Normalize the vector so that the total allocation remains the same

Example: [0, 100, 0, 100] -> [0, 105, 0, 90] -> [0, 107.7, 0, 92.3]

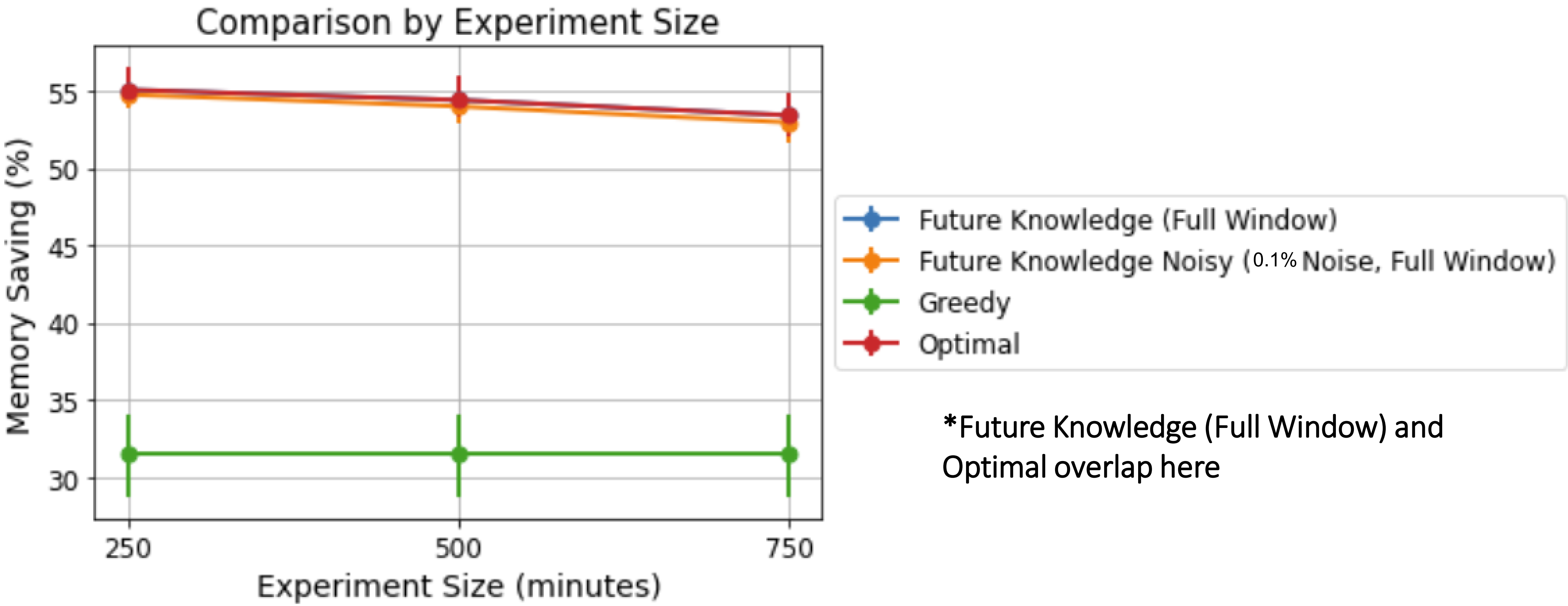
Improved Setup (Future Knowledge, No Migration)

This is the current setup, with a sliding window of a fixed size that advances by 5 minutes each time (very slow).

This is another possible setup, with disjoint intervals (much faster). But it’s less realistic.

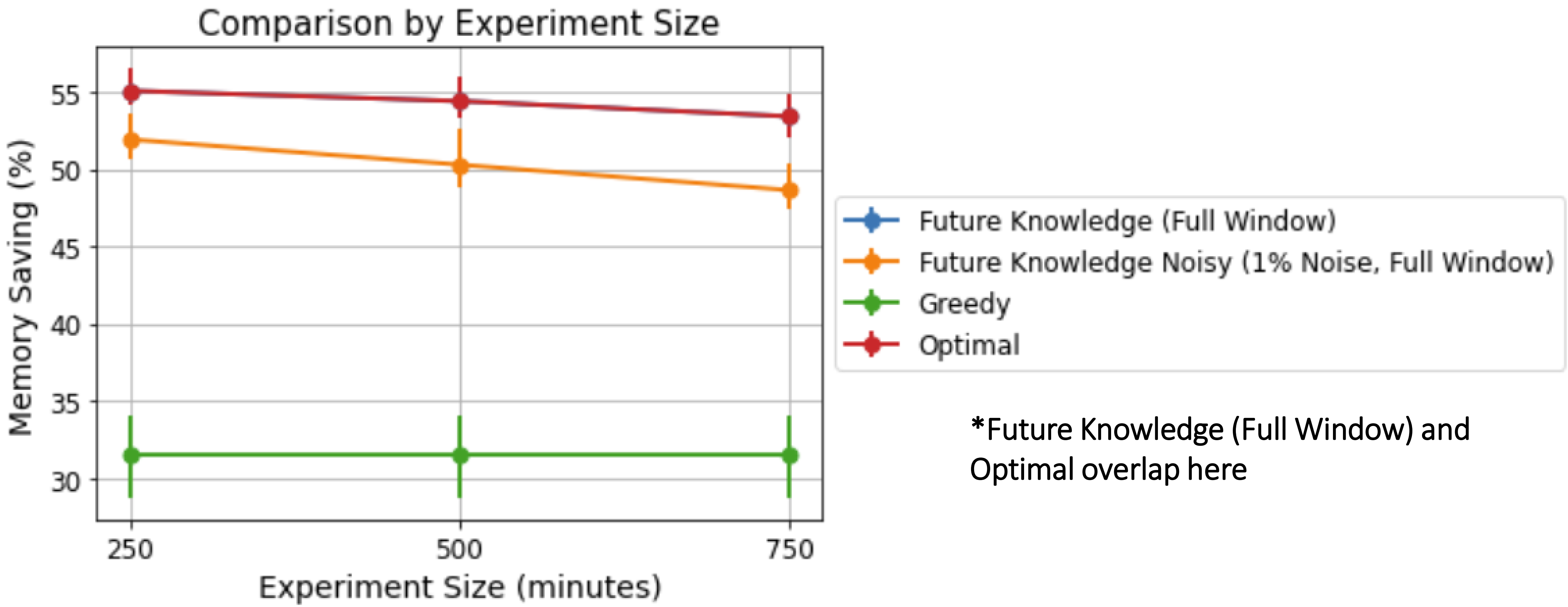
Without noise, the left one is more accurate, but if the noise gets compounded at each step, then which is better?

Noisy Future Knowledge



*Future Knowledge (Full Window) and Optimal overlap here

Noisy Future Knowledge



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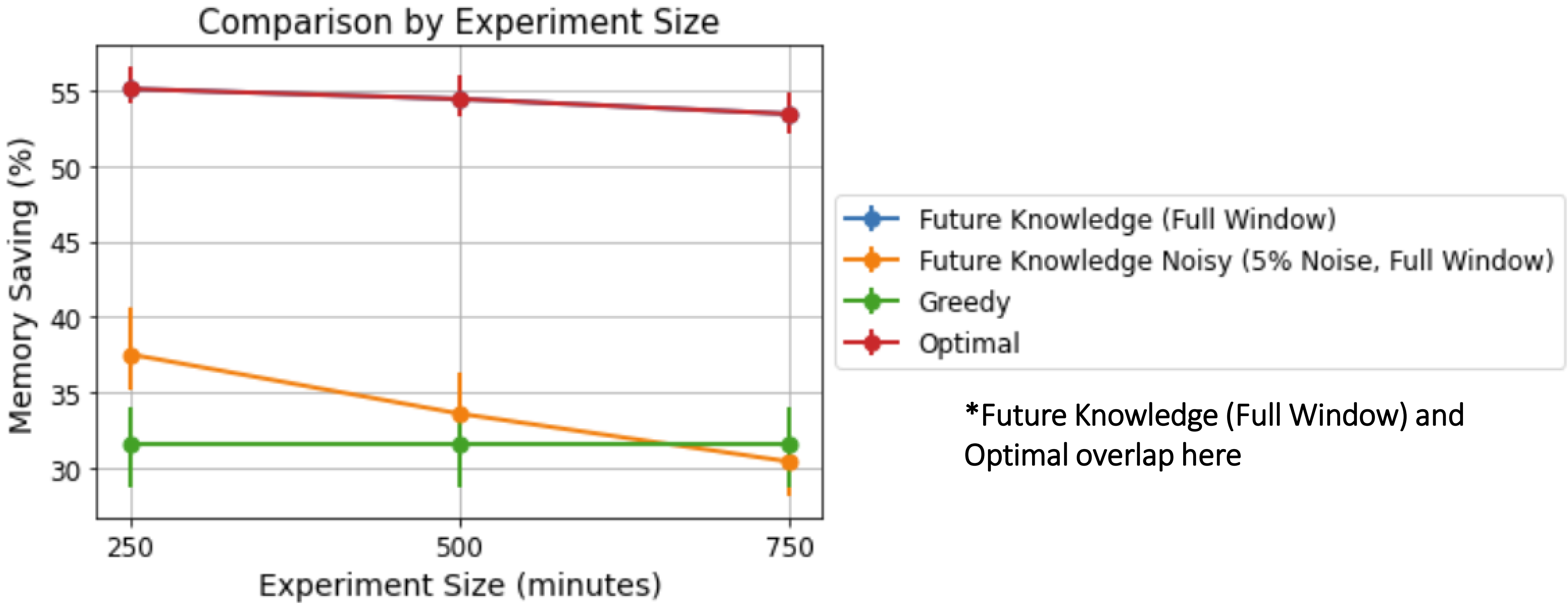
Noisy Future Knowledge



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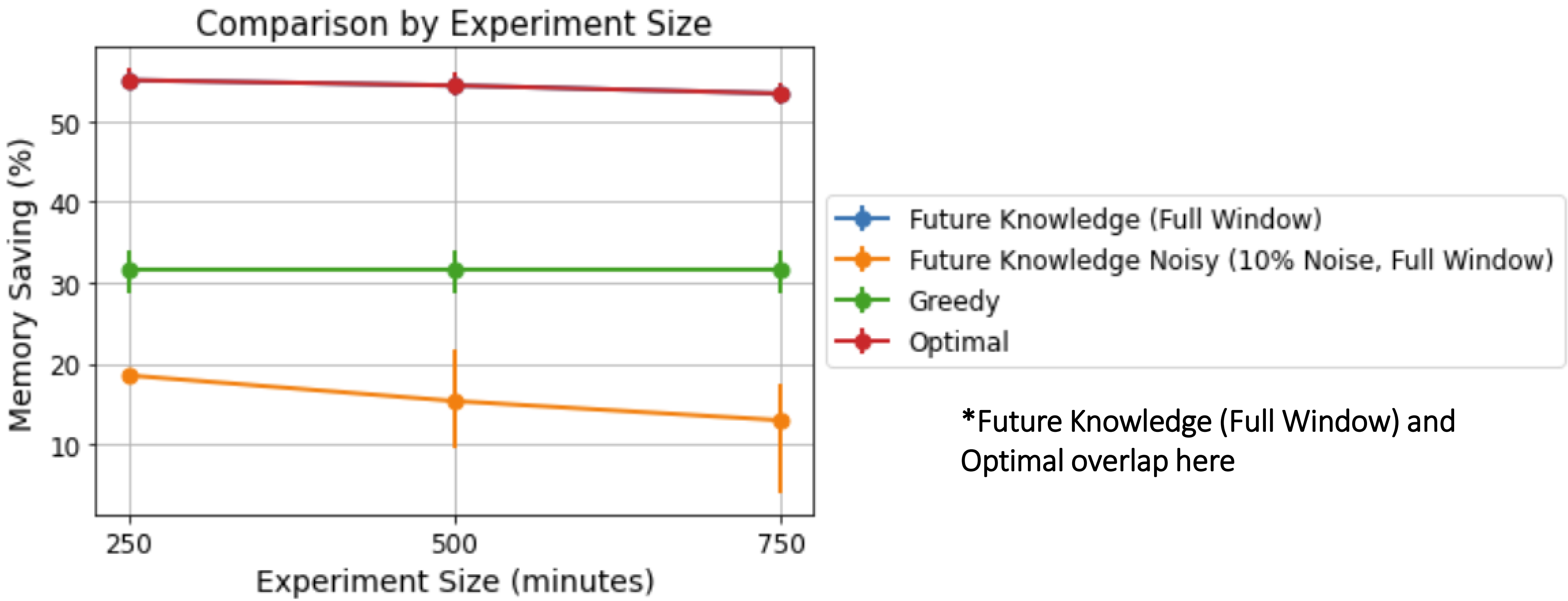

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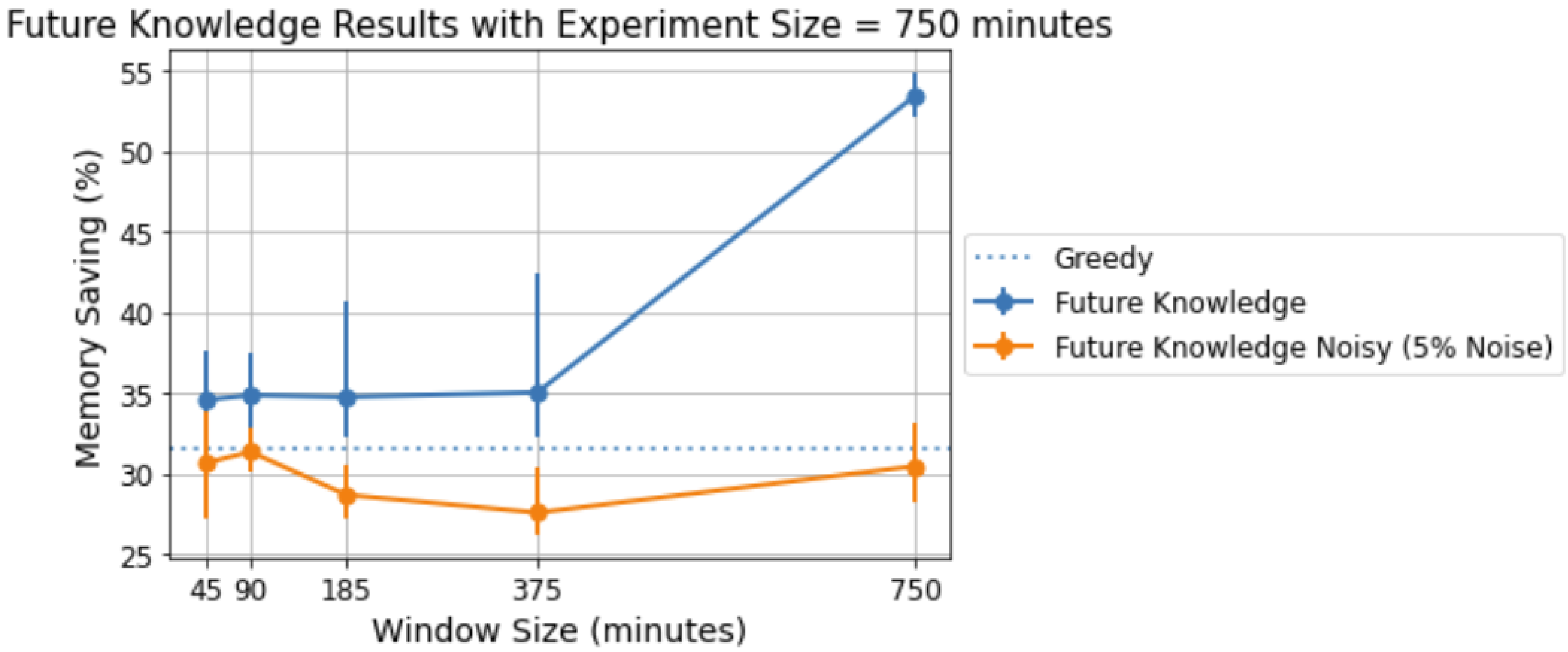

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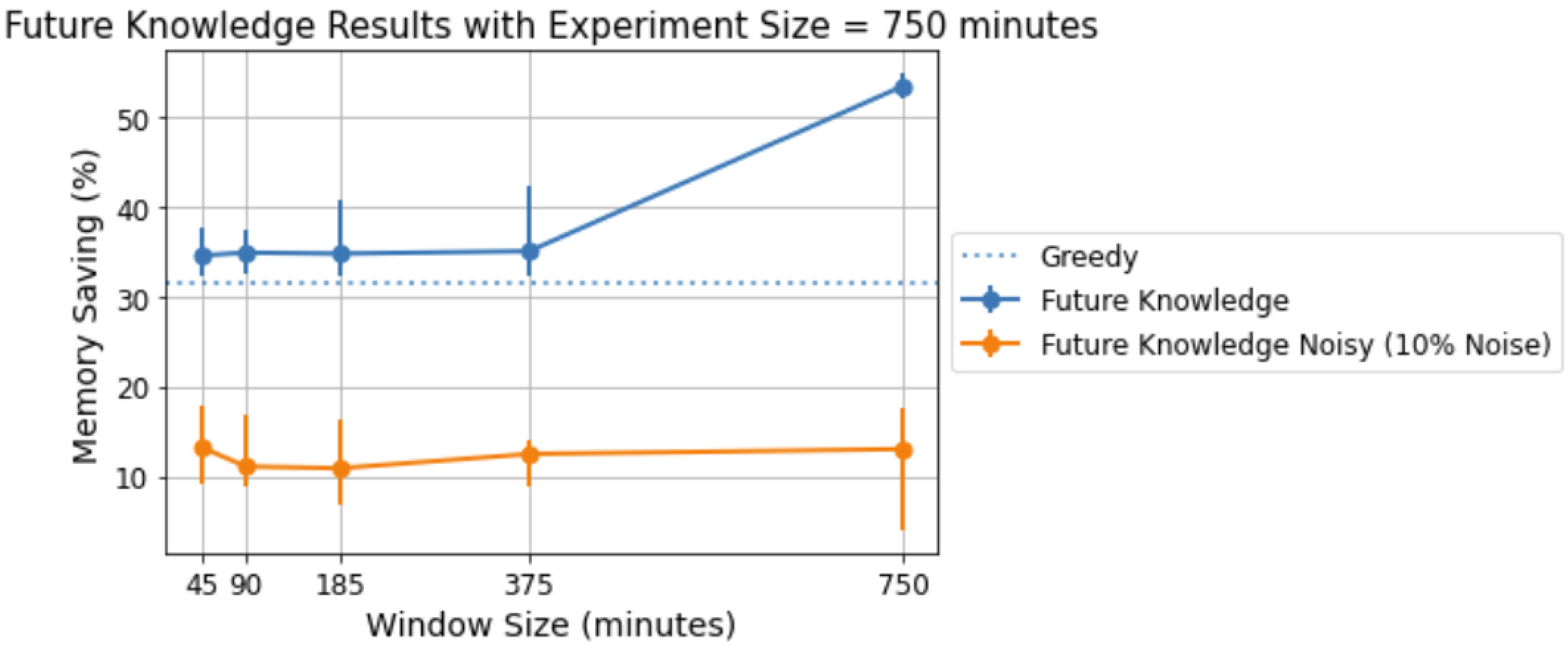

Conclusion

If at each step of the future knowledge algorithm every component of our allocation vector is off by at most even 5%, then in the long run we could actually do worse than greedy.

However, I think that because the sliding window only moves by 1 timestep (5 minutes) each time, then the noise gets compounded over an 8-day VM trace, so this result could be too pessimistic.

Should the disjoint sliding window be used?

Future Work

Trying out the disjoint sliding window.

Optimal value for how far out we should realistically be predicting.

Starting to do the predictions for the VM traces.

Sources

https://www.mdpi.com/1424-8220/19/22/5026 (sliding window diagrams)